

Attention Market Simulator

Applying Wall Street Math to Human Behavior

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The Unusual Idea

WHAT IF...

We treated human attention like a financial market?

FINANCIAL MARKETS have:

- Buyers and sellers (supply/demand)
- Price movements (volatility)
- Order flow imbalances
- Market regimes (bull/bear/sideways)

HUMAN ATTENTION has:

What I Built to Test This

A SIMULATION ENVIRONMENT:

- Models attention as a 'price' that rises and falls
- 6 behavioral regimes (engaged, bored, fatigued, etc.)
- Stochastic processes (random but realistic patterns)
- 2,000 timesteps of simulated behavior

35 TRADING-INSPIRED FEATURES:

- Order flow imbalance (notifications vs engagement)
- Rolling volatility (attention stability)
- Liquidity metrics (how easily attention shifts)

Key Concept: Order Flow Imbalance

IN STOCK TRADING:

- Order flow = buy orders vs sell orders
- Imbalance = more buyers than sellers (or vice versa)
- Predicts short-term price movements

IN ATTENTION MODELING:

- 'Buy pressure' = notifications, content, alerts
- 'Sell pressure' = declining engagement, boredom
- Imbalance = notifications vs actual attention given

Order flow imbalance is a real quantitative trading strategy used by hedge funds

Simulation Behavior Over Time

The simulation exhibited realistic patterns:

Phase 1 (0-500): High engagement → gradual decay

Phase 2 (500-1000): Volatility clustering (dopamine spikes)

Phase 3 (1000-1500): Disengagement → fatigue accumulation

Phase 4 (1500-2000): Stabilization at low baseline

Just like financial markets:

- ✓ Trending phases (engagement → disengagement)
- ✓ Volatility clustering (spikes grouped together)
- ✓ Mean reversion (returns to baseline)

Machine Learning Performance

REGIME
CLASSIFIER

75%

Accuracy

5-STEP
FORECASTING

$R^2 = 0.65$

65% variance

Q-LEARNING
AGENT

+15%

engagement

REGIME CLASSIFIER: Identifies which behavioral state you're in

FORECASTING: Predicts attention 5 steps ahead using market features

RL AGENT: Learns optimal notification policies (when to be aggressive/conservative)

Why This Cross-Domain Approach Matters

DEMONSTRATES TRANSFERABLE THINKING:

1. PATTERN RECOGNITION ACROSS DOMAINS

- Identified structural similarities between markets and behavior
- Applied quantitative trading concepts to psychology

2. PROVEN METHODS WORK IN NEW CONTEXTS

- Order flow imbalance (finance) → engagement prediction (behavior)
- Volatility clustering (stocks) → attention spikes (humans)

This is how quantitative researchers think: find universal patterns, apply them creatively

- Regime classification (trading) → behavioral states (psychology)

The 6 Behavioral Regimes

Similar to market regimes (bull, bear, sideways, crash):

1. **ENGAGED: High attention, sustainable engagement**
2. **BASELINE: Neutral state, moderate attention**
3. **DISENGAGED: Low attention, high boredom**
4. **FATIGUED: High attention but exhausted**
5. **OVERSTIMULATED: High volatility, unstable**
6. **ADDICTIVE LOOP: High attention + volatility + fatigue**

The Random Forest classifier identifies these regimes with 75% accuracy, enabling adaptive notification policies.

What This Project Demonstrates

- ✓ **QUANTITATIVE THINKING:** Applied market microstructure theory creatively
- ✓ **SIMULATION DESIGN:** Built stochastic environment with realistic dynamics
- ✓ **ML ENGINEERING:** Random Forest classifier + regressor, reinforcement learning
- ✓ **DOMAIN EXPERTISE:** Deep understanding of both finance and behavioral science
- ✓ **RESEARCH MINDSET:** Novel application, rigorous testing, validated with data

This is exactly how quantitative researchers work: find patterns, test rigorously, build predictive models